

Problem Statement

Intelligent Product Recommendation Chatbot: A Virtual Sales Associate for E-commerce

Overview

Online shopping platforms offer thousands of products across different categories, brands, price ranges, and specifications. While customers have access to a large variety of products, finding the most suitable product often becomes difficult.

Most e-commerce websites rely on keyword-based search and filtering systems. These systems work well when customers know exactly what they want, but they are less effective when customers have vague requirements, need guidance, or want expert recommendations.

For example, a customer may not search for a specific product but instead ask questions such as:

- Suggest some black shirts for my upcoming Goa trip.
- Recommend the best camera phone under ₹40,000.
- Compare iPhone 16 Pro and Samsung S25 Ultra.
- Help me buy a laptop for Computer Engineering studies.
- I am joining a gym next week. What shoes and clothes should I buy ?
- I am travelling to India next month. Which sunscreen should I purchase ?

These questions are difficult to answer using traditional search systems because they require understanding customer intent, preferences, budget, lifestyle, and use cases.

The goal of this project is to build an AI-powered conversational chatbot that acts as a virtual sales associate and helps customers discover, compare, and select products through natural language conversations.

Problem

Customers often face the following challenges while shopping online:

Too Many Choices

A single product category may contain hundreds or thousands of options, making it difficult to identify the best product.

Lack of Product Knowledge

Customers may not fully understand technical specifications and product features. For example:

- Which processor is suitable for programming ?
- Is 8GB RAM enough ?
- Which camera phone is better for photography ?

Difficulty Comparing Products

Customers frequently compare multiple products before making a purchase decision. However, manually reviewing specifications and features is time-consuming and confusing.

Unclear Requirements

Many customers know their goal but do not know the exact product they need. Examples:

- I want a phone for content creation.
- I need a laptop for college.
- I need comfortable shoes for daily walking.

These requirements must be translated into actual product recommendations.

Decision Fatigue

Too many options often lead to confusion, causing customers to delay or abandon purchases.

Proposed Solution

Develop an Intelligent Product Recommendation Chatbot that behaves like an experienced sales associate.

Instead of forcing customers to browse products manually, the chatbot should engage in a natural conversation, understand customer needs, and provide personalized recommendations.

The chatbot should be capable of:

- Understanding natural language questions.
- Identifying customer intent.
- Asking follow-up questions when information is missing.
- Recommending suitable products.
- Explaining why a product is recommended.
- Comparing products side-by-side.
- Suggesting alternatives within different budgets.
- Providing human-like shopping assistance.

Example Interactions

Example 1: Travel Shopping

- **User:** ‘I am going to Goa next month. Suggest some black shirts.’
- **Chatbot:** Understands that Goa is typically associated with warm weather and casual clothing and recommends suitable black shirts for travel and vacation use.

Example 2: Budget-Based Recommendation

- **User:** ‘Suggest the best camera phone under ₹40,000.’
- **Chatbot:** Finds phones within the budget and recommends options based on camera quality, battery life, and overall value.

Example 3: Product Comparison

- **User:** ‘Compare iPhone 16 Pro and Samsung S25 Ultra.’
- **Chatbot:** Provides a structured comparison covering Camera, Performance, Battery, Display, Software, Pricing, Pros and Cons.

Example 4: Need-Based Recommendation

- **User:** ‘I need a laptop for Computer Engineering.’

- **Chatbot:** Identifies common engineering requirements such as programming, virtualization, project work, and multitasking before recommending suitable laptops.

Example 5: Lifestyle-Based Recommendation

- **User:** ‘I am joining a gym next week.’
- **Chatbot:** Recommends a complete set of products such as running shoes, gym T-shirts, shorts, water bottle, and gym bag instead of recommending a single item.

Expected Outcome

The chatbot should make online shopping easier by helping customers:

- Discover products more efficiently.
- Understand product differences.
- Compare alternatives.
- Receive personalized recommendations.
- Make informed purchasing decisions.

Rather than acting as a simple search engine, the chatbot should function as a knowledgeable virtual sales associate that guides customers throughout their shopping journey using natural conversation. The overall objective is to create a smarter and more interactive product discovery experience that feels similar to receiving assistance from a human salesperson.

Project Scope

As part of this project, a realistic dummy e-commerce product dataset must be created or collected containing products, specifications, categories, pricing information, and product descriptions across multiple domains such as electronics, fashion, fitness, and personal care.

In addition to developing the chatbot logic, a complete end-to-end demo application should be built. This includes a user-facing frontend where customers can interact with the chatbot and a backend system that processes user queries, retrieves relevant products, performs comparisons, and generates recommendations. The final deliverable should demonstrate the complete customer journey from asking a product-related

question to receiving personalized recommendations and product comparisons through a conversational interface.